

K VAMSHI K

Data Engineer

☎ +1-571-748-3371 ✉ kvamshik10@gmail.com [in](#) [LinkedIn](#)

Professional Summary

Results-driven **Data Engineer** with 6+ years of experience designing and implementing scalable **data pipelines**, **cloud-based data platforms**, and **real-time analytics solutions**. Proficient in leveraging **Azure**, **AWS**, and **Snowflake** to optimize data processing, reduce costs, and improve decision-making. Expertise in **ETL/ELT workflows**, **data lake architecture**, and **dimensional data modeling**. Skilled in deploying **machine learning models** and building **interactive dashboards** for actionable insights. Adept at leading cloud migrations, automating workflows, and mentoring teams to drive productivity and innovation.

Technical Skills

Cloud Platforms: Azure (Databricks, Synapse Analytics, Data Lake Storage, Machine Learning, DevOps, Data Factory, SQL Database, Blob Storage), AWS (Redshift, S3, EMR, Lambda, Glue, Kinesis, CloudFormation, EC2, IAM, Athena, QuickSight), Snowflake

Data Engineering Tools: Apache Spark (PySpark, Spark Streaming), Azure Databricks, AWS Glue, Apache Kafka, Apache Airflow, Snowpipe, Delta Lake

ETL/ELT Tools: Azure Data Factory, AWS Glue, Informatica PowerCenter, Talend, Apache Airflow, Snowpipe

Programming Languages: Python, SQL, PL/SQL, PySpark, Shell Scripting, PowerShell

Big Data Technologies: Hadoop, Apache Spark, AWS EMR, Snowflake, Kafka, Delta Lake, Azure Data Lake Storage (ADLS Gen2)

Databases: Azure Synapse, Snowflake, Oracle, DB2, SQL Server, AWS Redshift, Hive, MySQL, PostgreSQL

Data Modeling: Star Schema, Snowflake Schema, Dimensional Modeling (Erwin)

Data Visualization: Power BI, Tableau

DevOps & CI/CD: Azure DevOps, Jenkins, CI/CD Pipelines

Scheduling Tools: Apache Airflow, Control-M

Data Security: Azure Security, AWS IAM, Encryption Strategies

Monitoring & Logging: Azure Monitor, AWS CloudWatch

Version Control: Git

Methodologies: Agile, Scrum, Data Pipeline Optimization, Performance Tuning, DataOps

Data Formats: Parquet, JSON, CSV, XML, Delta Tables

Performance Optimization: Query Optimization, Spark Tuning, Cluster-Level Optimization, Redshift Performance Tuning

Other Tools: Pandas, Jupyter Notebooks, Informatica MDM, Talend, Erwin

Professional Experience

JP Morgan Chase

Senior Data Engineer

May. 2023 – Present

Falls Church, Virginia

- Designed and deployed a real-time **fraud detection system** using **Azure Databricks** and **Spark Streaming**, processing **5M+ transactions/day** with **97.9% accuracy**, reducing fraudulent activities by **25%**.
- Led the migration of **10+ legacy ETL pipelines** from on-premise systems to **Azure Synapse Analytics**, reducing operational costs by **30%** and improving query performance by **40%**.
- Built a **data lake architecture** on **Azure Data Lake Storage (ADLS Gen2)** to consolidate **20+ data sources**, enabling seamless access to **15 TB** of structured and unstructured data for analytics teams.
- Developed **Python-based data pipelines** using **PySpark** to process and transform **1 PB+** of financial data, reducing processing time from **8 hours** to **2 hours**.
- Implemented **Snowflake** as the primary cloud data warehouse, optimizing query performance by **25%** and reducing storage costs by **20%** through efficient clustering and partitioning.
- Automated **CI/CD pipelines** using **Azure DevOps** and **Jenkins**, reducing deployment time by **50%** and improving release frequency to **2-3 times per week**.
- Designed and implemented **dimensional data models** (Star and Snowflake schemas) using **Erwin**, enabling seamless integration of data from **Oracle**, **DB2**, and **SQL Server**.
- Built real-time dashboards in **Power BI** and **Tableau** for executive reporting, providing actionable insights that improved decision-making for **\$500M+ revenue streams**.
- Collaborated with data scientists to deploy **machine learning models** on **Azure Machine Learning**, improving prediction accuracy by **20%** for customer churn analysis.

- Optimized **Kafka producers and consumers** to handle **10M+ messages/day**, reducing latency to under **2 seconds** and ensuring **99.5% uptime**.
- Designed and integrated a highly scalable Kafka-based data ingestion pipeline, processing over **100GB of data daily** with an **average latency of under 5 seconds** and a **peak throughput of 10,000 messages per second**.
- Created and optimized **Apache Spark clusters** on **Azure Databricks** to process **100 TB of data**, improving processing speed by 35% and enabling faster, more efficient data analytics and decision-making across the organization
- Reduced manual effort by **15 hours/week** by automating data ingestion and transformation tasks using **Snowpipe** and **Apache Airflow**.

Infosys

Jul. 2020 – Jul 2022

Data Engineer

Bengaluru, India

- Migrated **50+ ETL workflows** from **Informatica PowerCenter** to **AWS Glue**, reducing runtime by **35%** and cutting infrastructure costs by **20%**.
- Designed and deployed a **Big Data platform** on **AWS EMR** using **Hadoop, Hive, and Spark**, enabling the analysis of **1 PB+** of customer data for personalized marketing campaigns.
- Developed **Python scripts** for data cleansing and validation, improving data quality by **15%** and reducing manual errors by **90%**.
- Implemented a **Lambda architecture** using **Kafka** and **Spark Streaming** for real-time analytics, processing **5M+ events/day** with **99.5% uptime**.
- Created **CloudFormation templates** for infrastructure automation, reducing deployment time from **2 days** to **2 hours**.
- Integrated **AWS Redshift** with **Tableau** for business intelligence reporting, enabling stakeholders to access real-time insights and improving campaign ROI by **12%**.
- Built real-time data pipelines using **AWS Kinesis** and **Lambda**, processing **1M+ events/hour** for operational monitoring.
- Optimized **DynamoDB** performance by implementing efficient partitioning and indexing strategies, reducing query latency by **30%**.
- Conducted training sessions for junior engineers on **AWS services (EC2, S3, Lambda)** and **ETL best practices**, upskilling a team of **10+ members**.
- Designed and implemented a **data governance framework** using **Informatica MDM**, improving data consistency across **3+ systems**.
- Automated data validation processes using **Talend**, reducing manual effort by **10 hours/week** and improving data accuracy by **20%**.
- Built interactive dashboards in **Power BI** for customer segmentation, enabling targeted marketing campaigns that increased customer retention by **15%**.
- Reduced infrastructure costs by **25%** by optimizing **AWS EMR clusters** and implementing cost-effective storage strategies.

The Mavericks Technologies

Jul. 2018 – Jul. 2020

Data Engineer

Hyderabad, India

- Assisted in migrating **5+ ETL workflows** from on-premise systems to **Azure Data Factory**, reducing runtime by **30%** and cutting infrastructure costs by **20%**.
- Wrote and optimized complex **SQL queries (T-SQL, PL/SQL)** for data extraction and transformation, improving query performance by **25%** and reducing report generation time from **4 hours** to **1 hour**.
- Contributed to the design and implementation of a **data warehouse** using **Azure SQL Data Warehouse** (now Synapse Analytics), enabling faster analytics for **3+ business units** and reducing query latency by **40%**.
- Developed **Python scripts** and **PowerShell scripts** to automate data ingestion from on-premise systems to **Azure Blob Storage**, reducing manual effort by **10 hours/week**.
- Created interactive dashboards in **Power BI** for business intelligence reporting, providing actionable insights that improved decision-making for **\$100M+ revenue streams**.
- Assisted in implementing **data governance policies** using **Informatica MDM**, improving data consistency and accuracy across **2+ systems**.
- Played a key role in migrating **10+ TB** of historical data from on-premise **Hadoop clusters** to **Azure Data Lake Storage (ADLS Gen2)**, ensuring zero data loss and **99.9% data accuracy**.
- Monitored and troubleshooted **ETL pipelines** using **Control-M** and **Azure Monitor**, reducing downtime by **15%** and improving system reliability.
- Gained hands-on experience with **Azure services** such as **Azure Data Factory**, **Azure SQL Database**, and **Azure Blob Storage**, contributing to the successful delivery of **3+ cloud migration projects**.

Education

George Mason University

Fairfax, VA

Master of Science in Data Analytics Engineering;

Certifications

- Microsoft Certified Azure Data Engineer Associate
- Microsoft Certified Power BI Data Analyst Associate
- AWS Certified Data Engineer Associate